Adi Sharma

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Education

Sikkim Manipal Institute of Technology - Sikkim Manipal University **2020 – 2024**
Bachelor of Technology, Computer Science and Engineering *CGPA: 8.53 / 10*

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Structures, Design & Analysis of Algorithms, Distributed Systems, Operating Systems, Computer Networks, Parallel Programming, Theory of Computation, Discrete Mathematics

Industry Experience

Dell Technologies **Jan 2026 – Present**
Software Development Engineer 2 *Bengaluru, KA, India*

- Developed the **ITAssets Ops Copilot**, an LLM assistant on Llama 3.3 70B that lets IT support staff query employee, device and incident records and make configuration changes in plain English through 37 typed tools, replacing manual lookups across five separate consoles and cutting lookups per ticket from about five to under two.
- Wrapped deterministic checks around every model decision so the AI stays reviewable: tool arguments are validated against JSON schemas, column names are verified against the live database schema so the model cannot invent fields, permissions are enforced again at execution, and every write requires explicit user confirmation with the before-state captured in an audit log.
- Kept the service dependable with a primary and a fallback LLM endpoint, a circuit breaker so gateway outages never touch the main dashboard, and per request latency, token and cost logging; released through shadow mode and a gradual canary with automatic rollback.
- Grounded policy answers with retrieval over embedded IT policy and API documentation stored in PostgreSQL with pgvector, so the assistant answers from approved documents rather than model memory.
- Built the employee profile service in Node.js and TypeScript that merges seven upstream sources (Workday, Active Directory, ServiceNow, Intune, SCCM, Microsoft Graph, internal org API) in parallel and returns partial data instead of failing when a source is down, serving about 100K+ employees, with a rule engine of over 20 checks that surfaces context like expired passwords, new hires and VIP status in a single call instead of five separate tools.
- Designed the platform's four tier role based access control and access request workflow on PostgreSQL: Azure AD JWT authentication, safeguards against duplicate submissions and double approvals, email notifications isolated so a mail failure never blocks an approval, and an audit log entry on every permission change.
- Built a configuration driven SQL engine that lets partner teams expose new reporting endpoints by adding database configuration instead of code changes, with input sanitization and parameterized queries to prevent SQL injection, and migrated the platform's persistence from SQLite to PostgreSQL with a live data migration script.
- **Skills:** TypeScript, Node.js, Express, PostgreSQL, pgvector, SQLite, REST APIs, LLM tool calling, RAG, LLM, Azure AD OAuth2, SSO, HashiCorp Vault, Kubernetes

Dell Technologies **Aug 2024 – Dec 2025**
Software Development Engineer 1 *Bengaluru, KA, India*

- Built the real time command and control plane for the **Software Removal platform** on SignalR and WebSockets, with a RabbitMQ queue as a durable second channel for guaranteed delivery, supporting both fleet wide broadcast and machine targeted dispatch with per device success reporting, managing over 100,000 Windows devices across three global regions at a 99.7% success rate.
- Paired it with a lightweight .NET Windows service agent that receives commands over both channels, executes removals on each device and streams results and logs back in real time, cutting install size from 350 MB to 30 MB and installation time from 10 minutes to 30 seconds.
- Built the software title normalization engine behind the same platform, matching 16,000 messy vendor titles across 8 GB of collected inventory evidence using fuzzy string matching and TF-IDF similarity at 75% accuracy, raising processing throughput from 2,400 to 16,000 records per hour.
- Architected the FastAPI microservice for Dell Communique that turns natural language briefs into structured email templates through an internal LLM gateway with OAuth2 token refresh, recovering malformed model output with a layered JSON parser and fallback, lifting parse success from roughly 85% to 98% and cutting time to first token from about 4 seconds to 200 milliseconds with streaming.
- Productionized Go analytics services for Communique that parallelized three PostgreSQL report aggregations from about 25 seconds down to 6 seconds, and built its predictive layer with a survival model flagging users likely to disengage.
- Built a reinforcement learning scheduler that learns each device's activity patterns and places inventory scans in idle windows, reaching 95% idle period execution and lowering average CPU impact from 28% to 6%.
- **Skills:** Python, Go, C#, .NET 8.0, SignalR, FastAPI, XGBoost, scikit-learn, PyTorch, Sentence-BERT, MLflow, PostgreSQL, Redis, RabbitMQ, Docker, Kubernetes, GitLab CI/CD

Dell Technologies **Jan 2024 – May 2024**
Software Development Engineer Intern (Winter) *Bengaluru, KA, India*

- Developed a semantic search engine over Elasticsearch using Sentence-BERT embeddings in a hybrid keyword and dense retrieval setup, fine tuned on 12,000 training queries, raising user satisfaction from 58 to 84% and bringing queries per successful search from 8 to 2.3.
- Built a hybrid recommendation system pairing matrix factorization with content based filtering over TF-IDF and metadata, improving click through rate from 23% to 41%, with MLflow tracking the A/B experiments.

- Architected the training request workflow engine for Dell Digital People in Node.js, GraphQL and PostgreSQL as a state machine task model, and rebuilt the CI/CD pipeline to take build success from 58% to 94% and build time from 45 minutes to 12 minutes.

Dell Technologies **May 2023 – Jul 2023**
Software Development Engineer Intern (Summer) *Bengaluru, KA, India*

- Engineered the User360 dashboard on FastAPI and React for the TMX Data Platform, bringing API latency from 3.2 seconds to 0.4 seconds through query plan analysis, Redis caching at a 76% hit rate and connection pooling.
- Optimized retrieval over 500,000 records from 45 seconds to 1.8 seconds using table partitioning, cursor based pagination and materialized views on PostgreSQL, and scaled API throughput from 50 to 340 requests per second through indexing.
- Architected the vendor budget ingestion pipeline in Express with Redis and a background worker, acknowledging multi-MB Excel uploads in under 500 ms, then validating and sanitizing every row against injection attacks before cascading upserts across four relational tables.

Research Experience

Birla Institute of Technology and Science, Pilani **Jun 2022 – Aug 2022**
Summer Research Intern, under Prof. Dr. Hari Om Bansal *Pilani, RJ, India*

- Developed an AI based coordination system for a heterogeneous robot fleet of rovers and drones contending for limited charging infrastructure, formulating autonomous resource management as a joint forecasting and assignment problem under hard energy constraints.
- Designed an LSTM power load forecasting model over telemetry time series reaching above 90% accuracy with mean absolute error under 5 W, giving reliable battery drain prediction across varying mission profiles.
- Implemented the Hungarian algorithm for optimal task assignment coupled with dynamic charging optimization, reaching a 92% mission completion rate and 72% charging port utilization, and validated the full pipeline in ROS and Gazebo with real time telemetry.
- **Skills:** Python, TensorFlow, LSTM, ROS, Gazebo, Combinatorial Optimization, Time Series Modelling, Linux

Technical Skills

Languages: Python, JavaScript / TypeScript, Go, C#, SQL, Bash
AI and LLM Engineering: LLM integration and tool calling, prompt engineering, structured outputs, RAG, embeddings and vector search (pgvector, Sentence-BERT), OCR and document extraction, golden dataset and LLM-as-judge evaluation, hallucination detection, model fallback and routing, PII masking, LangChain, LangGraph
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, MLflow
Backend and Data: FastAPI, Node.js, Express, React, ASP.NET Core, REST APIs, OAuth2, PostgreSQL, SQLite, MongoDB, Redis, RabbitMQ, Elasticsearch
Engineering Practices: Git, GitLab CI/CD, GitHub Actions, Docker, Kubernetes, unit and regression testing, observability and monitoring, canary releases and rollback, Linux, Azure

Projects

Documentation Q&A Assistant using Retrieval Augmented Generation | *Python, LangChain, FastAPI, pgvector, Node.js* **2025**

- Built a question answering assistant over 600 pages of public product and API documentation, with an ingestion pipeline that parses PDF and HTML sources, falls back to Tesseract OCR for scanned pages, and splits text along section headings into about 4,800 indexed chunks in PostgreSQL with pgvector.
- Orchestrated answering as a four step LangGraph flow (retrieve, draft, verify, answer) served from FastAPI, with a React and Express front end that stores sessions, shows the source passages behind every answer, and collects helpful or not helpful feedback per response.
- Measured every retrieval change against 180 hand labelled questions with known source sections: moving from fixed size chunks and plain vector search to heading aware chunking, combined keyword and vector search, and a reranking step took Recall@5 from 0.71 to 0.89.
- Cut unsupported claims from 14% to 4% with a verification step that checks each generated sentence against the retrieved passages and declines to answer when support is missing, scoring changes with an LLM judge over three repeated runs to separate real gains from run to run noise.
- Wired the evaluation into GitHub Actions so a prompt or retrieval change fails the build on regression, tracked latency and cost per answer, and masked emails and personal identifiers with a regex and named entity pass before any text reached the model.

DDoS Detection in Software Defined Networks using ML | *Python, scikit-learn, POX, Mininet, OpenFlow* **2024**

- Developed an intrusion detection system for software defined networks, with a real time feature extraction pipeline over OpenFlow statistics deriving 23 flow level features through sliding window aggregation at 89 ms detection latency.
- Handled class imbalance with SMOTE and cost sensitive learning, then compared five classifiers across attack types; Random Forest performed best at 98.9% accuracy and recall after a grid search over 180 hyperparameter configurations.
- Integrated the inference engine with the POX controller through an asynchronous event driven architecture with batched inference, taking throughput from 15 to 560 flows per second, with 110 ms mitigation time through dynamic flow rule installation.

Image Captioning System for Assistive Technology | *TensorFlow, Keras, ResNet50, LSTM, Flask* **2023**

- Designed an end to end captioning pipeline pairing pre-trained CNN vision encoders (ResNet50, VGG16) with LSTM decoders to generate natural language image descriptions for visually impaired users.
- Compared four encoder and decoder combinations under BLEU based evaluation, reaching BLEU-1 of 0.61 with the ResNet50 and Bi-LSTM pairing, and deployed the model as a Flask application with text to speech, returning spoken captions for uploaded images in real time.

Honors & Awards

Dell Innovation Award (2025): for the real time SignalR agent POC and window-service behind the Software Governance System managing devices globally at dell

Dell Extraordinary Award (2025, twice): for the Software Inventory Agent deployed globally across 100,000 devices, which reduced unauthorized software titles from 4,100 to 1,800.

Dell Bravo Award (2026 and 2025): for the ITAssets Ops platform with 4-tier RBAC Backend Services development, and for MS sharepoint integration delivered on tight timelines for Dell Communique.

Dean's List, Sikkim Manipal University (2023) and **Birla Education Trust Academic Excellence Scholarship** of Rs. 25,000, both awarded for academic merit.

Dell Hack-2-Hire Hackathon Winner (2022): built a computer vision and language generation model converting images into spoken descriptions as assistive technology for visually impaired users.

Second Position, IETE Innovation Challenge (2019): Institution of Electronics and Telecommunication Engineers, for an autonomous FPV rover for ranging and surveillance.

Teaching, Mentoring & Service

Dell Development Program, Committee Lead (2026) Conducting Technical and community events, mentoring and supporting new graduate hires at Dell.

Google Developer Student Club, Lead (2022 to 2023): selected as one of 560 GDSC Leads across India to lead Sikkim's sole Google DSC, organising coding workshops, study jams and tech talks for over 500 students, and invited to the GDSC Leads Conference.

CS Student Council, General Secretary (2023): delivered technical workshops for over 100 students on Computer Networks, Data Structures and Algorithms, and led council outreach teaching over 50 underprivileged school students.

AI Club, ML Mentor (2021 to 2022): mentored over 50 students in machine learning fundamentals through hands-on TensorFlow, scikit-learn and PyTorch sessions.

Google Developer Group Bangalore, Core Member and Event Organizer (2024 to Present): organised and hosted developer meetups end to end as emcee, with over 200 attendees per event.

Technical leadership and service: Head of IT at TEDxSMIT, Head of IT at IAESTE SMU, Tech Team Lead for Smart India Hackathon, and NSS volunteer for the 2023 Sikkim flash flood relief effort.

IPR Filed

IPR 1: On-Device Privacy-Preserving Prompt Refinement Using Cluster-Conditioned Local Agents — On-device small language model prompt compiler using a typed intermediate representation and a 17-operation edit DSL, with federated learning based on differentially private, noised rule-outcome histograms and two companion filings. *Dell Technologies, 2026.*

IPR 2: PSF-Based Display Pre-Compensation with Integrated Wavefront Sensing and Vision-Aware Biometric Authentication — Display pre-compensation system integrating point spread function (PSF) modeling, wavefront sensing, and vision-aware biometric authentication. *Dell Technologies, 2026.*

IPR 3: Behavioral Graph Query Disambiguation — Graph-based approach for resolving ambiguity in user queries using behavioral context and interaction signals. *Dell Technologies, 2026.*

IPR 4: Agentic AI System for Software Removal Governance — Agentic AI framework for software removal governance combining XGBoost-based risk prediction, K-Means persona classification, and Isolation Forest anomaly detection. *Dell Technologies, 2025.*